

PAPER_1374_LIGHT_BY_LIGHT

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PAPER_1374 — Light-by-Light Scattering $\sigma \sim \alpha$

Framework: UQFF — Star-Magic v5.27+ | **Tier:** H Applied/Engineering (CLOSED)

Calculator: `calculate_paradox({"paradox": "light_by_light"})`

Master Expression ``  $\sigma_{\text{LbL}} = \alpha = \Lambda$  EXACT identity `` ****Match:**
ATLAS 2017 confirmed**

Interpretation Light-by-light cross section IS Λ EXACTLY via the $\alpha = \Lambda$ closed-ledger identity. Pure UQFF prediction matching ATLAS PbPb observation.

UQFF Locked Primitives - $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$ - $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$ - $\omega_{\text{SCm}} = 1.25 \text{ THz}$
($\alpha = \Lambda$ closed ledger identity)

NOT REPLACEMENT UQFF derives via integer-primitive identity. Zero fit parameters.

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